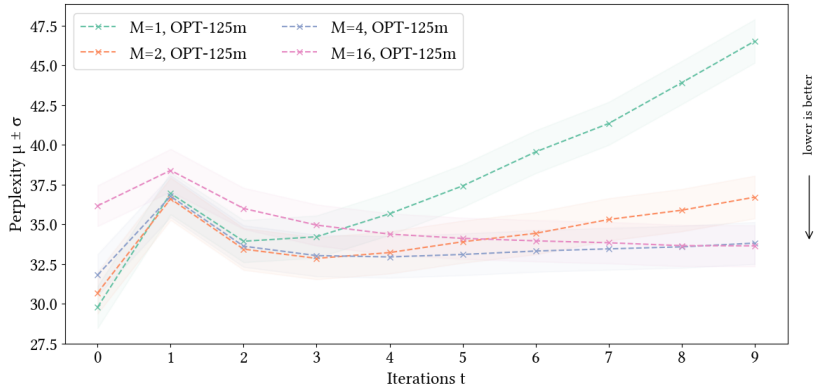
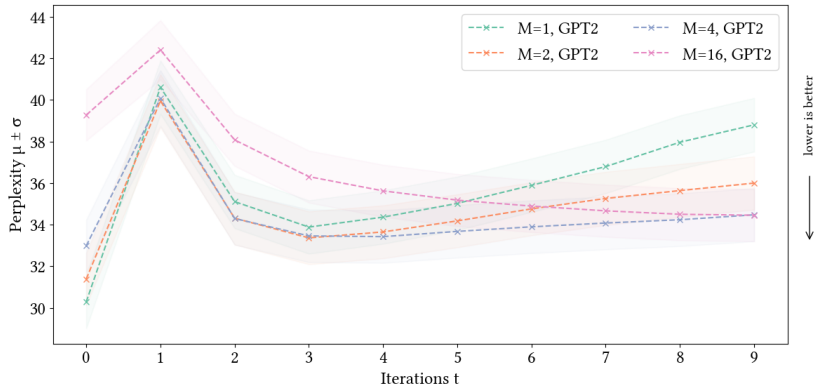
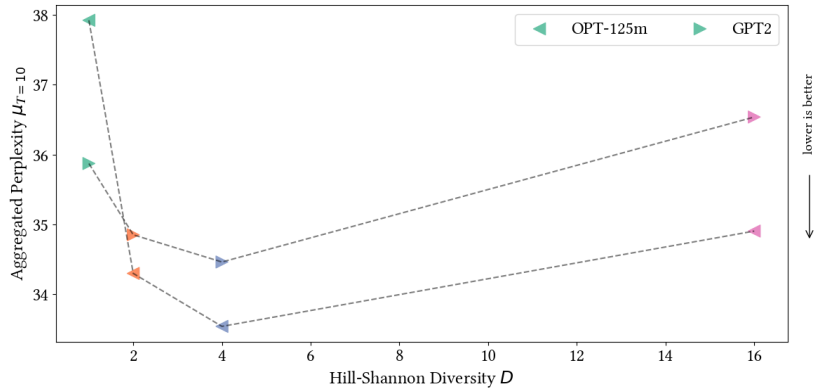


(a) OPT-125m

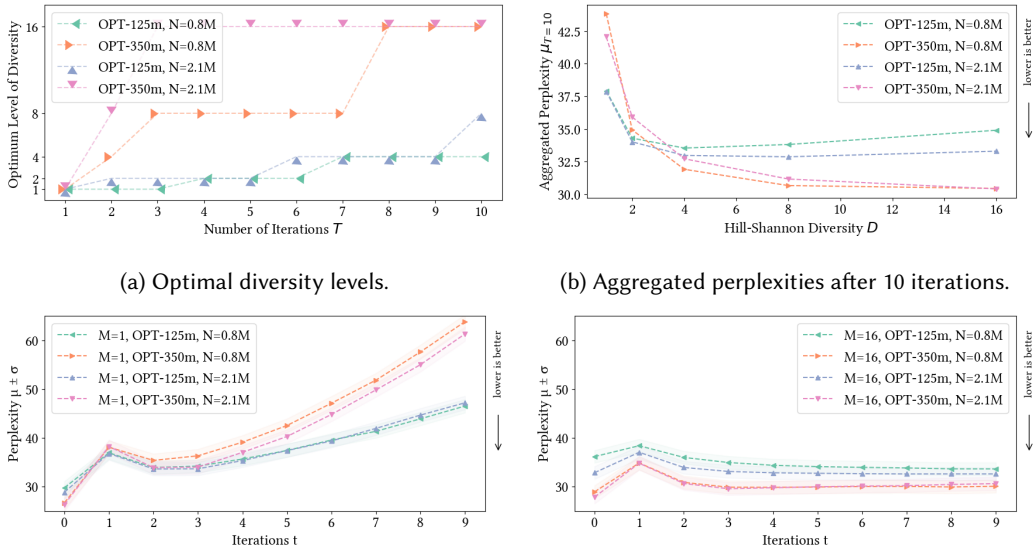


(b) GPT2



(c) Aggregated perplexity after 10 iterations

Fig. 4. **Ecosystem diversity mitigates collapse.** (a, b) Comparison of ecosystems' mean perplexity with increasing diversity ($D = M$) at fixed dataset size across ten iteration steps. While low-diversity ecosystems ($D = 1, 2$) exhibit strong performance decay, the high-diversity ecosystem ($D = 16$) starts with relatively high perplexity but improves over t . The shaded region is a combined standard deviation across the models and data points. (c) Aggregated perplexity. Both model families exhibit an optimal level of diversity of 4 for overall performance across 10 iterations. Colors correspond to the diversity levels shown in Figure 2 (e.g., orange denotes $M; D = 2$).



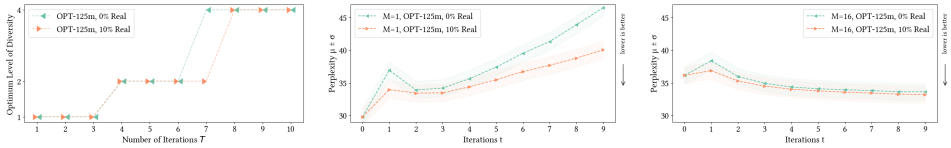
(a) Optimal diversity levels.

(b) Aggregated perplexities after 10 iterations.

(c) Mean perplexities for M;D=1

(d) Mean perplexities for M;D=16

Fig. 5. **The optimal level of diversity increase with model and dataset scale.** Ecosystem diversity across four system scales (two model sizes × two dataset sizes; variation V1). "OPT-125M, $N = 0.8$M" (green ◀) corresponds to the primary setting, using a 125M-parameter model and a total training dataset of 0.8M tokens. **Top:** (a) Optimal diversity level vs. number of iterations T and (b) aggregated perplexity $\mu_{T=10}$ for five diversity levels ($M; D = 1, 2, 4, 8, 16$). Larger models (orange and pink vs. green and blue) show a steeper increase in optimal diversity with T, shifting the optimum from $D = 4$ to $D = 16$ after 10 iterations. Increasing dataset size (pink ▼ and blue ▲ vs. green and orange) mainly affects larger models, increasing the optimal diversity level when considering fewer than 8 iterations. **Bottom:** Perplexity trajectories across ten iterations for two diversity levels (c: $D = 1$, d: $D = 16$). For a fixed diversity level, increasing model or dataset size reduces perplexity at $t = 0$. (c) In the most homogeneous ecosystem ($D = 1$), larger models show strong perplexity increases in later iterations indicating stronger collapse, while solely increasing dataset size has only a modest effect. (d) In highly diverse ecosystems ($D = 16$), larger models achieve lower perplexity across all iterations, while increasing dataset size benefits primarily the smaller model. The shaded region represents the combined standard deviation across models and data points.

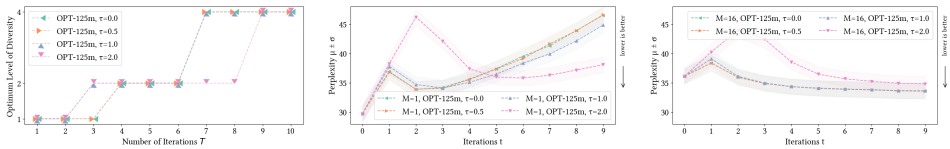


(a) Optimal diversity levels.

(b) M;D=1

(c) M;D=4

Fig. 6. **Replacing 10% of artificial data with real data has only a minor effect on the optimal diversity level**. 0% real data (green ◀) corresponds to the primary setting. (a) Optimal diversity level vs. number of iterations T. (b, c) Perplexity trajectories across ten iterations for two diversity levels (b: $D = 1$, c: $D = 16$). For a fixed diversity level, replacing artificial data with real data at each iteration reduces perplexity, while differences across ecosystems diminish as diversity increases.



(a) Optimal diversity levels. (b) M;D=1 (c) M;D=16

Fig. 7. **Increasing sampling temperature ($\tau = 0.0, 0.5, 1.0, 2.0$) has only a minor effect on the optimal diversity level.** $\tau = 0.0$ (green ◀) corresponds to the primary setting. (a) Optimal diversity level vs. number of iterations T. (b, c) Perplexity trajectories across ten iterations for two diversity levels (b: $D = 1$, c: $D = 16$). A sampling temperature of $\tau = 1.0$ (blue ▲) slightly reduces perplexity in the most homogeneous ecosystem ($D = 1$) and has almost no effect in the most diverse setting ($D = 16$). For both diversity levels, $\tau = 2.0$ (pink ▼) causes a sharp perplexity increase in early iterations, with a relative decrease at later iterations only for $D = 1$.

less pronounced, two patterns emerge: increasing dataset size for the the larger model improves performance primarily in the most homogeneous ecosystem ($D = 1$), whereas for the smaller model it improves performance primarily in the most diverse setting ($D = 16$).

4.2 V2: Real-data incorporation and sampling temperature

Improving data quality by incorporating 10% real data (Figure 6) or by applying temperature sampling with $\tau = 1.0$ and $\tau = 2.0$ (Figure 7) generally improves model performance. Because both interventions primarily benefit the most homogeneous ecosystem ($D = 1$), performance differences across diversity levels shrink relative to the primary setting. However, since the positive effect of ecosystem diversity exceeds these improvements, both interventions produce a similar trajectory of the optimal diversity level over iterations as observed in the primary setting. We highlight several notable performance effects relative to the primary setting below.

The impact of incorporating real data at each iteration increases as diversity decreases, reflecting the larger per-model dataset size ($n = N/M$). Temperature sampling can also affect data quality by diversifying generated outputs. For temperatures of $\tau = 1.0$ and $\tau = 2.0$ in the most homogeneous ecosystem ($D = 1$), perplexity increases during early iterations but declines by a similar magnitude later in the loop, resulting in only marginal overall improvement. In more diverse ecosystems, $\tau = 1.0$ produces slightly negative effects, while $\tau = 2.0$ leads to strongly negative effects. By contrast, a temperature of $\tau = 0.5$ shows almost no effect relative to the primary setting for any fixed diversity level.

5 Theoretical interpretation

Across experiments, our study of ecosystem evolution reveals a monotonic increase of the optimal diversity level with number of self-training iterations T. For a fixed number of 10 iterations, we find a U-shaped relationship between diversity and collapse. Considering that dataset size per model $n = N/M$ decreases as diversity increases, the observed increase in optimum diversity level indicates a complex interplay between opposing effects of diversity and scale with T. The complexity grows when considering the results obtained with increased scale (Section 4.1): When model size increases, we observe stronger collapse in low-diversity ecosystems ($D \leq 2$) and weaker collapse in high-diversity ecosystems ($D \geq 4$). Increasing dataset size generally improves performance for OPT-125m but worsens performance for OPT-350m, except in the $D = 1$ ecosystem. In this section we propose a theoretical explanation of the underlying effects.